Given a value V, you have an infinite supply of N different coin denominations. Find the minimum number of coins required to make exactly V cents.

You must also print the coins used in an optimal solution.

If it is not possible to make the value V, print -1.

Input Format

The first line contains an integer T, the number of test cases. For each test case: The first line contains two integers V and N. V — the target value. N — the number of available coin denominations. The second line contains N integers C[i], representing the coin denominations.

Constraints

1 ≤ T ≤ 100 1 ≤ V ≤ 10^6 1 ≤ N ≤ 10^6 1 ≤ C[i] ≤ 10^6

Output Format

For each test case:

Print the coins used in a minimum-coin solution, separated by spaces. In the next line, print the minimum number of coins. If it is impossible to make V, print -1.

Sample Input 0

```
7 2
2 4
```

Sample Output 0

```
-1
```

Sample Input 1

```
11 4
1 5 6 9
```

Sample Output 1

```
2
```

```python
V, N = map(int, input().split())
coins = list(map(int, input().split()))

dp = [999999] * (V + 1)

dp[0] = 0

for i in range(1, V + 1):

    for coin in coins:

        if coin <= i:
            dp[i] = min(dp[i], dp[i - coin] + 1

if dp[V] == 999999:
    print(-1)
else:
    print(dp[V])

```

Line: 19 Col: 1

Upload Code as File

Run Code Submit Code

Test against custom input

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

Sample Test case 0

Sample Test case 1

Input (stdin) Download

```
7 2
2 4
```

Your Output (stdout)

```
-1
```

Expected Output Download

```
-1
```